Using Novel Hybrid Convolutional Neural Network for Dysarthria Diagnosis

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***Abstract*— Dysarthria is a motor speech disorder characterized by articulation and phonation difficulties resulting from speech muscle weakness, paralysis, or incoordination. A precise and timely diagnosis of dysarthria is essential for effective treatment and management of the condition, as it may deteriorate over time or be a precursor to a much more serious disease. On the other hand, this is becoming a severe problem in recent times owing to the rising ageing population and the prevalence of neurological disorders among such people. Hence early detection of dysarthria is deemed essential for the timely management of the disease. In recent years, Artificial Intelligence (AI) applications have shown promising results in various audio processing tasks and incorporated into pathological voice analysis for disease diagnosis. The majority of previous studies on dysarthria detection employed Machine Learning (ML) and Deep Learning (DL) models as the disease classification models. In light of this, this study presents a novel hybrid approach for classifying dysarthria based on audio data using Convolutional Neural Network (CNN) and Support Vector Machine (SVM). According to the experimental results, the proposed classification schema achieves an accuracy of 98.25 % compared to previous research work. Overall, our proposed method aims to automate the classification process, enabling faster and more reliable diagnoses.**

Keywords—Artificial Intelligence, Convolutional Neural Network, Dysarthria, Speech Classification, Machine Learning, Deep Learning, SVM

I. INTRODUCTION

Dysarthria is a common speech disorder caused by neurological conditions such as stroke, parkinson disease, and cerebral palsy [1]. According to [1], it is estimated that 30% of the global population suffers from dysarthria. The severity of dysarthria is directly proportional to the amount of neuromuscular impairment present. Patients who suffer from dysarthria will, in most cases, have trouble correctly pronouncing words and enunciating them clearly [1]. In extreme circumstances, individuals will exhibit symptoms such as aphasia and even have trouble breathing [1]-[3].

Traditional methods for assessing dysarthria rely heavily on the subjective judgment and intrusive diagnosis of medical professionals; therefore, there is a great deal of subjective variation among clinicians. Consequently, traditional methods are challenging, time-consuming, and arduous, and their results are highly subjective [1],[2]. As of result, pathological voice recognition technology has evolved to provide a more objective foundation for medical diagnosis and treatment [1]-[3]. This technology can distinguish between normal voices and the abnormal voices produced by dysarthria patients.

In recent years, the study of pathological speech analysis and recognition has garnered considerable interest from both society and the industry. In contrast, manual classification of dysarthria based on audio recordings is time-consuming and susceptible to subjective interpretations [1]-[3]. As a remedy, using an AI-based recognition method, dysarthria patients can be distinguished from normal individuals through feature extraction of problematic voices and the development of a recognition model. When this information is paired with pertinent pathological knowledge, a further evaluation of the severity of the ailment may be performed. Apart from ML, deep neural networks, also known as DNN, have lately come to the attention of worldwide researchers who are interested in delving further into the investigation of anomalous voices / pathological voice recognition [3],[4]. This is mostly attributable to the DNN's greater potential for learning features and then performing predictions. The majority of the techniques that are being used now involve learning from classic features, such as the Mel-Frequency cepstrum coefficient (MFCC) [2]-[6].

In recent years, there has been a clear growth in interest in applying CNNs and other techniques that are based on DL for the purpose of automating the classification process [1]-[4]. Such techniques yield faster, and more reliable diagnoses as opposed to traditional pathological voice recognition methods. Thus, motivated by the fact that discovering more efficient AI-enabled methods for dysarthria diagnosis, in the study, we present a novel hybrid CNN and SVM integrated classification method based on the obtained MFCC features.

The primary goal of this research is to successfully differentiate normal persons from those suffering from dysarthria in terms of their voices by first extracting the necessary statistical information from particular disordered sounds, then building a recognition model with the essential algorithms, and then realizing that differentiation. The building of a recognition model is dependent on both the traditional approaches to machine learning (SVM) and the more recent deep learning approaches (CNN). In this regard,

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the main contributions of the study are described in the following.

- Present a novel hybrid classification schema for dysarthria diagnosis integrating deep learning and machine learning approaches towards achieving higher accuracy.
- Provide a summary of state-of-the-art, summarizing what classification methods have been used with their contributions to differentiate our work from theirs.
- Obtained results are compared with state-of-the-art to validate the results we obtained.

As this research is a part of the ASEAN IVO 2023 project, "Spoof Detection for Automatic Speaker Verification," which aims to distinguish a genuine voice from a spoofed voice, this research presents a novel AI approach for classifying dysarthria voice over normal voice. The remainder of the paper is organized as follows. Following the introduction, the second section presents a state-of-the-art in pathological voice analysis and dysarthria classification. Section three presents the methodology of the research and the designed model. Following section three, section four highlights the results obtained from the experimental work along with our observations. Finally, conclusions derived from the research are highlighted in section five.

II. LITERATURE REVIEW

Dysarthria refers to a group of speech disorders caused by paralysis, a reduction in the intensity of muscular contractions, or incoordination in the movement of speech-related muscles as a direct consequence of neuropathy [1]-[5]. The primary symptoms of dysarthria include the inability to form words or sounds, uneven rhymes, a sluggish flow of language, and rhythm difficulties [1],[2],[6]-[10].Pathological speech recognition is one of the useful supplementary methods that is used to do objective evaluation of dysarthria assessment [7]-[10].

In general, speech-language pathologists (SLPs) are typically trained to conduct dysarthric evaluations in a clinical setting. During clinical examinations, SLPs typically attempt to estimate and classify intelligibility based on the number of words understood [11-[15]. However, this auditory-based method is subjective and can be affected by factors such as assessor familiarity and the use of various rating scales. These issues introduce errors and make assessments unreliable, impeding the efficacy of treatment [11]-[15]. The manual nature of these evaluations, on the other hand, makes them costly and inaccessible to individuals with limited access to SLPs and speech clinics. Therefore, automated intelligibility assessment methods offer a cost-effective, uncomplicated, and less subjective method of classifying dysarthric speakers and reducing errors [8]-[11].

By implementing automated intelligibility assessment approaches, it is possible to enhance assessment performance and facilitate remote treatment, recovery plans, and rehabilitation for dysarthric speakers [15]-[18]. Alternatively, while traditional computation methods may fall short in capturing and learning the complexities of human speech, deep learning algorithms offer a promising avenue for developing predictive models [5]-[8], thereby significantly improving dysarthric intelligibility assessments. Common ML methods include SVM, Artificial Neural Network (ANN), Decision Tree (DT), and Bayesian Network (BN), according to the current state of the art [2]-[6]. DL methods, meanwhile, included CNN, Recurrent Neural Network (RNN), and Transfer Learning [2]-[6]. Nonetheless, these DL techniques may establish an end-to-end multi-level recognition model utilizing various convolution kernels that can autonomously extract more valuable deep features, thereby improving the accuracy of classification or prediction compared to ML approaches.

In terms of the use of ML for pathological voice analysis ; (Vadovsky & Paralic, 2017), [19] used Random Forest and Classification and Regression Tree (CART) to investigate the most accurate classification model using speech signals. Overall, their highest accurate models gave an accuracy of 66.50 %. (Kadiri & Alku, 2020), [20] conducts a thorough examination of glottal source characteristics and explores their efficiency in voice pathology identification using SVM. (Amara et al., 2016),[21] provide an efficient method for voice pathology diagnosis based on speech signal processing and machine learning techniques, in which they employed SVM and Gaussian Mixture Model (GMM) with generated MFCC characteristics. (Vikram and Umarani, 2013) described a novel technique for pathological speech classification that combines Gaussian Mixture Model-Universal Background Model (GMM-UBM) and SVM in [22]. Overall, their method provided a 96.61% accuracy.

When it comes to DL for pathological voice analysis; (Wu et al., 2018),[23] introduced a unique approach for pathological voice identification that leverages spectrograms as the major characteristics derived from diseased and normal speech recordings. (Hire et al., 2022) describe an ensemble of CNNs for detecting parkinson disease in voice recordings from healthy persons and speech recordings from the PC-GITA, a publicly accessible database [24]. (Lee & Choi, 2020) offer a CNN and feed-forward neural network model for pathological voice classification that incorporates MFCCs, linear predictive cepstral coefficients, and higher-order statistics [25]. (Zhang et al., 2017) applied DNN for pathological speech classification, and their results shown that the DNN technique can successfully enhance the recognition rate for pathological voice recognition [26]. (Hung et al., 2022) presented a set of learnable Sinc functions to replace the very first layer of a frequently used CNN in order to create an explainable SincNet system for categorizing or identifying abnormal voices in [27]. They demonstrated that the suggested technique may enhance accuracy by almost 7% to 8% when compared to the baseline CNN model.

Having provided a brief overview of the use of ML and DL in pathological voice analysis, the next section provides a summary of recent research on dysarthria diagnosis.

A. *Summary of recent research*

To have a better understanding of the use of various AI-enabled techniques in the context of dysarthria diagnosis the following Table 1 summarizes the latest research that has been done.

TABLE 1 SUMMARY OF RECENT RELATED RESEARCH

Reference	Classifier/technique used	Acoustic features used	Contributions
(Ye et al., 2022) , [1]	Combination of 1DCNN and double-	MFCC	The authors have designed a post-stroke

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	LSTM (long short-term memory) networks		dysarthria detection model integrating 1DCNN and two LSTM networks.
(Joshy & Rajan, 2021), [3]	DNN, CNN and LSTM network	MFCC	The authors employed DNN, CNN, and LSTM networks to classify dysarthria severity using MFCC characteristics. They employed the UA-Speech corpus and the TORGO database for the study, and the performance of the aforementioned models is compared to the baseline SVM classifier.
(Dumane et al., 2021), [4]	CNN	MFCC	The authors propose a CNN model for identifying whether or not an individual has dysarthria. In order to examine the speech signals, the authors used zero crossing rates, MFCCs, spectral centroids, and spectral roll-off.
(Joshy & Rajan, 2023) , [5]	Residual convolutional neural network (RCNN)	Mel spectrograms	The researchers used RCNN and mel spectrograms to explore the efficacy of using the multi-head attention mechanism (MHA) and the multi-task learning technique for automated dysarthria severity level categorization.
(Mani Sekhar et al., 2022) , [6]	Transfer learning CNN	Mel spectrograms	The authors have designed a transfer learning-based CNN model for dysarthria speech classification. As per the learning features, they have used coverted mel spectrograms from the TORGO dataset.
(Hall et al., 2023) , [11]	CNN	MFCC	In the study, the researchers evaluated the efficacy of several DL-based automated dysarthric intelligibility evaluation models in terms of their ability to generalize to new speakers. In order to accomplish this, they exploited CNN models and employed MFCC as the features.
(Kadi et al., 2016) , [12]	Gaussian Mixture Model (GMM), SVM and hybrid GMM/SVM	MFCC	The researchers devised a dysarthria-specific automated speaker identification method to improve dysarthria analysis and categorization. Secondly, they proposed a method for automated dysarthria severity evaluation and used Nemours and Torgo datasets with MFCC characteristics.
(Sidi Yakoub et al., 2020) , [13]	CNN	MFCC	The researchers employed empirical mode decomposition and Hurst-based mode selection (EMDH) in conjunction with CNN to improve the recognition of dysarthric speech using the collected MFCC characteristics.
(Shih et al., 2022), [18]	CNN with Gated Recurrent Unit	MFCC	This study proposes an integrated CNN-GRU model for detecting dysarthria using convolutional neural networks and gated recurrent units.
Our work	CNN + SVM	MFCC	Our work combines CNN and SVM and designs a hybrid classification model for dysarthria detection in an efficient way.

III. METHODOLOGY

Towards employing a novel classification schema for dysarthria diagnosis, the methodology followed in the study comprised five steps, as depicted in Figure 1.

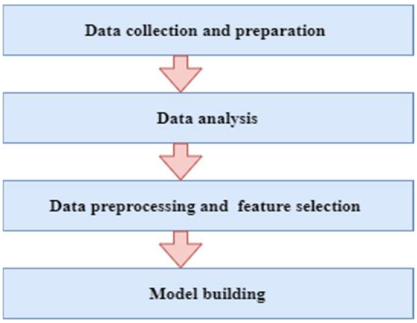


Fig. 1. Methodology followed.

As depicted in the first phase of Figure 1, the dataset was extracted and prepared from the TORGO database [7],[8]. The database includes male and female articulations, dysarthric articulations with aligned acoustics, and measured 3D articulatory features [7, 8]. The recordings were created by individuals with and without dysarthria to identify the concealed articulatory features through statistical pattern recognition. The.wav files in the data set contain both females and males with dysarthria as well as individuals without dysarthria. Overall, the dataset comprises 2,000 recording samples from individuals with and without dysarthria (1000 dysarthria samples and 1000 non- dysarthria samples).

A. Data analysis

As a result of the fact that people who have dysarthria pronounce words more slowly and with a pitch that is less consistent than healthy people, it is possible to see fluctuations in the amplitude using a waveplot. This is due to the fact that dysarthria affects people's ability to produce sound. The waveplot of the dysarthria patient displayed in Figure 2 is, in general, more unpredictable and exhibits a slurred than those given for the healthy individual, as depicted in Figure 2.

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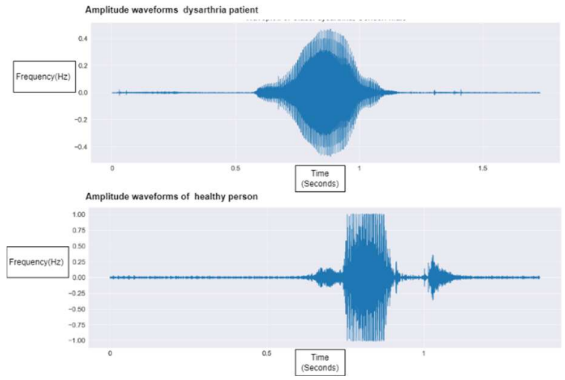


Fig. 2. Waveplots pertaining to dysarthria and non-dysarthria subjects

As depicted in Figure 3, the spectrograms of the dysarthria samples reveal that the energy magnitudes of the frequencies are more dispersed over time. This may either be interpreted as a sign of sluggish and slurred speech or as a result of the fact that the words are coming out more quickly and overlapping with one another. On the other hand, for the non-dysarthric samples, it is clear to observe that the energy magnitudes are more focused on the specific areas of the recording when the individuals have uttered anything at a regular speed. For obtaining the wave plots and spectrograms, the Librosa library in Python was used.

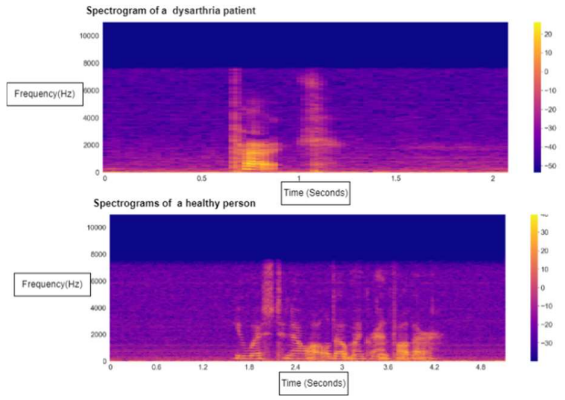


Fig. 3. Waveplots pertaining to dysarthria and non-dysarthria subjects

B. *Data pre-processing and feature selection*

Overall, the human ear is capable of picking up a wide range of tone frequencies, which are measured on a scale called Mel. In recent years for speech recognition tasks, the Mel-frequency cepstral coefficients, often known as MFCC, are highly utilized [11],[12]. As part of the data pre-processing and feature selection, the Librosa library in Python was used to extract the feature of the inverse coefficient of the mel frequency from the audio samples. The following Figure showcases the respective voice sample and mel spectrogram in Figure 4.

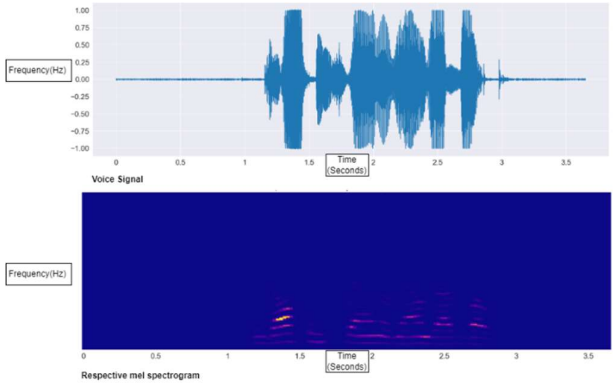


Fig. 4. Respective voice sample and mel spectrogram

Upon extracting the features, the data are then divided further, with 80% of data being used for model training and 20% being used for testing.

C. *Proposed hybrid CNN model*

First, the first CNN input layer of the model receives pre-processed data as input. For each audio file, the data was separated with the shape (16, 8, 1); thus, the first CNN with 32 filters had an input of (16 ×8 × 32) with the kernel (3 × 3). Following the input CNN layer, there is a max pooling layer, another CNN layer, and another max pooling layer. After all these layers, there was flatten layer, and afterward, there was a dense layer and SVM linear layer for making the binary classification. The detailed summary of layers and parameters of the proposed model are highlighted in Table 2. The overall architecture of the model is presented in Fig. 5

TABLE 2 SUMMARY OF LAYERS AND PARAMETERS OF THE PROPOSED MODEL

Network	Layer	Number of activations	No of parameters
CNN + SVM	Conv 1	(None, 16, 8, 32)	320
	Max pooling 1	(None, 8, 4, 32)	0
	Conv 2	(None, 8, 4, 64)	18496
	Max pooling 2	(None, 8, 4, 64)	0
	Flatten	(None, 512)	0
	Dense	(None, 64)	32832
	Dense + SVM	(None, 1)	65

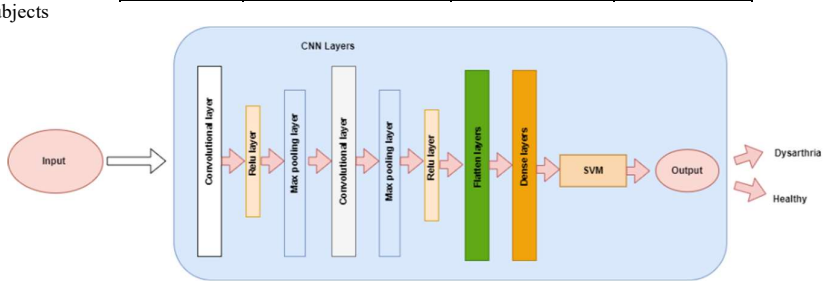


Fig. 5. The architecture of the proposed model

Table 3 presents the hyperparameters we have used to get the optimal results.

TABLE 3 HYPERPARAMETERS PERTAINING TO THE PROPOSED MODEL.

Parameters	Value
Optimizer	Adam
Epochs	28 (early stopping call back used)
Loss function	Hinge loss

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Learning rate	0.001
Batch size	32

IV. RESULTS AND DISCUSSION

Upon completing the model building, the effectiveness of the proposed model was evaluated using accuracy, recall, precision, and ROC (Receiver Operating Characteristic) curve, evaluation metrics.

After five epochs, the training loss decreases and demonstrates a linear loss, despite some fluctuations. The call-back function was implemented to terminate model training once the validation set reached the minimum loss threshold. After 28 training epochs, the training accuracy and testing accuracy reached 99.50 % and 98.25% consecutively. Thus, it can be concluded that our proposed model identifies the distinctive vocal data characteristics required to classify the supplied data as belonging to dysarthria or non-dysarthria patients. Figure 6 illustrates how the training and validation accuracy varies with the number of epochs, whereas Figure 7 illustrates how the training and validation loss varies with the number of epochs.

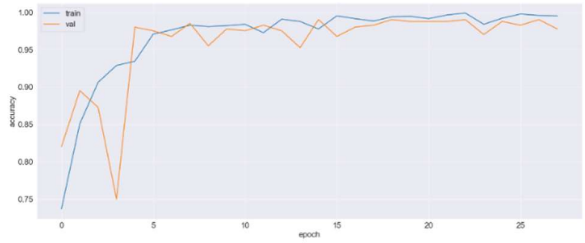


Fig. 6. Training and validation accuracy vs epochs

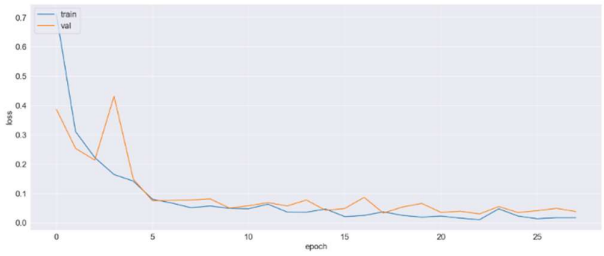


Fig. 7. Training and validation loss vs epochs

The following Figure 8 showcases the Area Under Curve (AUC) for our proposed model, which is almost 99%. This indicates that our model classification results are excellent.

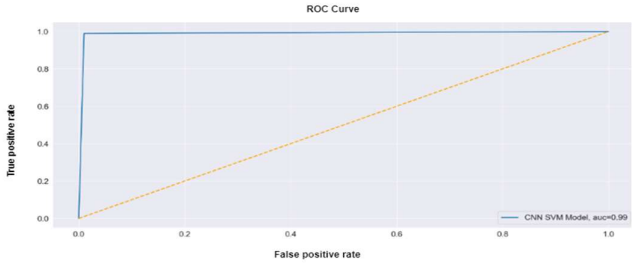


Fig. 8. ROC curve for the proposed model

The performance evaluation metrics of our proposed model are shown in Table 4.

Parameters	Accuracy	Precision	Recall	AUC ROC
Values (%)	98.25	97.50	96.50	99.00

TABLE 4 PERFORMANCE EVALUATION CRITERIA FOR OUR MODEL

TABLE 5 CONFUSION MATRIX

Actual/Prediction	Dysarthria	Non-dysarthria
Dysarthria	198	2
Non-dysarthria	2	198

According to the formulated confusion matrix highlighted in Table 5, it can be observed that out of 400 random audio samples, 396 audio samples were correctly predicted, and four audio samples were not correctly predicted. Table 6 provides a summary of the comparison between our research findings and those of other methodologies employed in previous studies. The researchers in [1], have used 1DCNN and double LSTM networks with their own dataset and achieved an accuracy of 97.40%. The researchers in [3] tried with varied DL and CNN classifiers and obtained an accuracy of 96.18% based on the speech data in the TORGO database. Accordingly, researchers in [4] and [6] used CNN as their classification model and managed to obtain 93.87% and 97.73% accuracy consecutively based on the speech data from the TORGO database. The researchers in [5] and [11] experimented with the UA-Speech database and managed to obtain an accuracy of 95.75% and 97.19% accuracy consecutively. Compared to all these models, our hybrid model yields an overall accuracy of 98.25%, based on the speech data from the TORGO database.

TABLE 6 PERFORMANCE COMPARISON

Reference	Classifier	Dataset	Accuracy
(Ye et al., 2022) ,[1]	1DCNN + double-LSTM	A syllable pronunciations database collected from Mandarin-speaking participants	97.40 %
(Joshy & Rajan, 2021) ,[3]	DNN, CNN, and LSTM network	TORGO database	96.18%
(Dumane et al., 2021), [4]	CNN	TORGO database	93.87%
(Joshy & Rajan, 2023) , [5]	Residual convolutional neural network	UA-Speech database	95.75
(Mani Sekhar et al., 2022) ,[6]	Transfer learning CNN	TORGO database	97.73%
(Hall et al., 2023) , [11]	CNN	UA-Speech database	97.19%
Our proposed model	CNN +SVM	TORGO database	98.25%

V. CONCLUSION

Dysarthria, a motor speech disorder, can pose challenges in accurate and timely diagnosis, affecting the quality of life for individuals affected by the condition. The proposed hybrid CNN model in the research study leverages the power of DL

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and combines it with other computational techniques (ML) to achieve improved accuracy in dysarthria detection; compared to the earlier methods employed in previous studies. Thus, by employing a hybrid CNN, researchers and healthcare practitioners can benefit from the complementary strengths of different algorithms and methodologies. The CNN's ability to extract intricate features from speech signals, coupled with other techniques that capture temporal dynamics or contextual information, results in a more comprehensive and discriminative analysis of dysarthric speech patterns. Therefore, the integration of our output model in the diagnosis process can enhance the efficiency and automation of assessments, making it a valuable tool for early detection and intervention. Overall, timely identification of dysarthria can lead to better treatment outcomes and personalized therapy plans tailored to each individual's specific needs. The future directions of the research include enhancing our proposed model capabilities and exploring new avenues for improved diagnosis and management by expanding the model's applicability by incorporating additional data sources, such as video analysis or physiological data, to provide a more comprehensive view of dysarthria. Nonetheless, efforts should be made to develop user-friendly interfaces that seamlessly integrate AI-assisted dysarthria detection into clinical practice. In conclusion, the novel hybrid CNN present in the study represents a promising step forward in dysarthria diagnosis, offering accurate, efficient, and data-driven solutions to improve the lives of individuals affected by dysarthria.

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